05. Speed

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EXERCISE 27

1. Complete the following table.

	Distance	Time	Speed
(a)	150 km	3 h	$\frac{150}{3}$ =
(b)	450 km	6 h	$\frac{450}{6}$ = 75
(c)		2 h	45 km/h
(d)	330 km		60 km/h

EXERCISE 28

1. A van traveled 216 km in 4 hours. Find its average speed in km/h.

2. David swam 800 m in 20 minutes. Find his average speed in m/min.

3. A car traveled at an average speed of 80 km/h for 5 hours. What was the distance traveled?

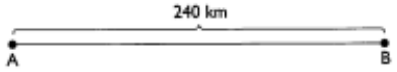
400 km

4. Jerome took 15 minutes to walk from his house to a community center. His average speed was 60 m/min. Find the distance between Jerome's house and the community center.

$$\begin{array}{r} 60 \\ \times\ 15 \\ \hline 300 \\ 60 \\ \hline 900 \end{array}$$

5. A train traveled 30 mi at an average speed of 60 mi/h. Find the time taken for the trip.

6. A car traveled from Town A to Town B at an average speed of 80 km/h. How long did the trip take?



EXERCISE 29

1. Shawn takes 2 minutes to cycle 800 m.
   (a) Find his average speed.
   (b) If he continues to cycle for another 5 minutes at the same average speed, how much further will he travel? Give your answer in kilometers.

$\frac{800}{2}$

a 400 m/min

b 2 km

$5 \cdot 400 = 2000$ m

2. Tyler took 2 hours to drive from Town A to Town B which were 150 km apart.
   (a) Find his average speed.
   (b) How many hours would he take to drive from Town B to Town A if the average speed for the return trip was 60 km/h?

$\frac{150}{2}$ = 75

a 75 kmph

$\frac{150}{60}$ = 2.5 hrs

b

3. Miguel walked 910 m from home to his office. His average speed was 70 m/min. He left home at 7:30 a.m. At what time did he arrive at his office?

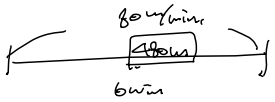
13

910

70

7:43

4. Amanda took 6 minutes to walk from her school to the nearest train station. Her average speed was 80 m/min. How long would she take if her average speed was 60 m/min?



80
× 6
480

$t = \frac{D}{v}$

$t = \frac{480}{60} = 8$ min

480
60

EXERCISE 30

1. Meihua jogged for $\frac{1}{2}$ hour at 6 km/h. She jogged for another $\frac{1}{4}$ hour at 8 km/h. What distance did she jog altogether?

45 min

$\frac{1}{2}$ 6kmph

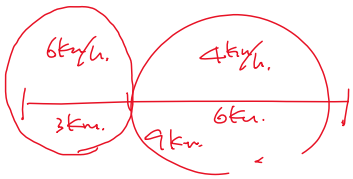
$\frac{1}{2} \times 6 + \frac{1}{4} \times 8$

2. In a 9-km walkathon, Susan walked at an average speed of 6 km/h for the first 3 km and at an average speed of 4 km/h for the remaining distance. How much time did she take for the walkathon?

6km

[redacted]m

6km

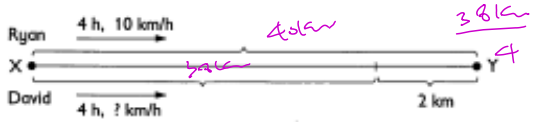


total

$t = \frac{D}{v}$ $\frac{3}{6}$ $\frac{6}{4}$ $\frac{3}{2}$

$\frac{1}{2}$hour + $\frac{3}{2}$hr ⇒ $\boxed{2\text{hours}}$

3. Ryan and David both cycled from Town X to Town Y. They started at the same time. After cycling for 4 hours, Ryan completed the trip but David still had 2 km to cycle. If Ryan's average speed was 10 km/h, find David's average speed for the 4 hours.



40km

38km / 4 = 9.5 km/h

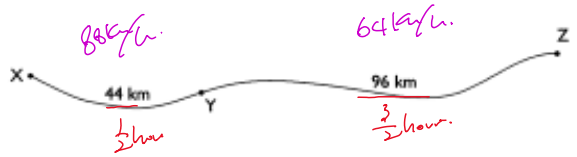
40 km

Distance traveled by David in 4 h = 38 km

$\boxed{9.5\ \text{km/h}}$

$4\overline{)38}$ = 9.5

4.



A car traveled from Town X to Town Y at an average speed of 88 km/h. Then it traveled from Town Y to Town Z at an average speed of 64 km/h.

(a) Find the total time taken.

$\frac{1}{2} + \frac{96}{64}$ $\frac{12}{8}$ $\frac{3}{2}$

1.5

2 hrs

$\frac{1}{2} = 30$ min

$\frac{96}{64}$ $\frac{6}{4}$ 2

(b) Find the average speed for the whole trip.

$88 + 64 = 152$

$152 \div 2 = 140$

$44 + 96 = 140$

$\frac{44+96}{2}$ km/hour

$22 + 48$

70 km/h

5. Daniel took 1 hour to drive from Town A to Town B at an average speed of 70 km/h. Then he took 2 hours to drive from Town B to Town C at an average speed of 64 km/h.

(a) Find the total distance traveled.

70
+128
198

198 km
~~66~~

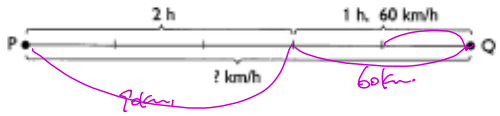
(b) Find the average speed for the whole trip.

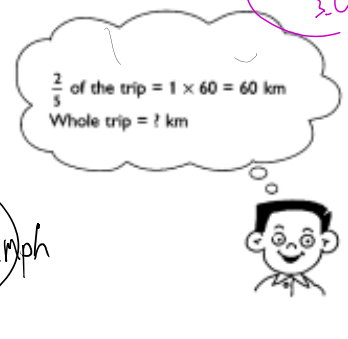
66
198
3

66 km/h

EXERCISE 31

1. Nicholas drove from Town P to Town Q. He took 2 hours to travel $\frac{3}{5}$ of the trip. He took another 1 hour to travel the remaining trip at an average speed of 60 km/h. Find his average speed for the whole trip.





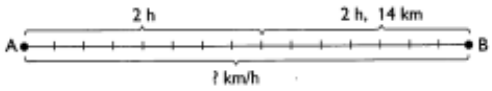
2. Ali cycled from Town A to Town B. He covered $\frac{1}{5}$ of the trip in the first hour and $\frac{1}{3}$ of the trip in the second hour. He took 2 hours to cycle the remaining 14 km. Find his average speed for the whole trip.

$\frac{5}{15}$ $\frac{3}{15}$

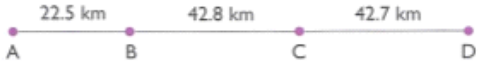
$\frac{1}{5} + \frac{1}{3} = \frac{3}{15} + \frac{5}{15}$

$= \frac{8}{15}$

He covered $\frac{8}{15}$ of the trip in 2 h.



1. The diagram shows the route of a bus from Town A to Town D.



The bus left Town A at 2:30 p.m. and arrived at Town D at 4:30 p.m. Find its average speed for the whole trip.

2. Danny took 2 hours to cycle from Town A to Town B. His average speed was 10 km/h.
   (a) Find the distance between the two towns.
   (b) If his average speed was increased by 2 km/h, how much time would he take for the trip?

3. Ben and David both cycled a distance of 24 km. They started at the same time. David completed the trip 20 minutes earlier than Ben. If David's average speed was 9 km/h, find Ben's average speed.

4. Paul took 5 minutes to walk from his house to the main road at an average speed of 46 m/min. He then took another 10 minutes to walk to his office at 40 m/min. How far did he walk altogether?

5. A group of boys went on a 12-km hike. For the first 3 km, they walked at an average speed of 4 km/h. For the rest of the hike, they walked at an average speed of 6 km/h. Find the total time taken.

6. Andrew drove from Town X to Town Y. He took 2 hours to travel $\frac{4}{5}$ of the trip. He took 1 hour to travel the remaining 30 km.
   (a) Find the distance between the two towns.
   (b) Find his average speed for the whole trip.

7. A motorist traveled from Town P to Town Q. After traveling $\frac{1}{3}$ of the trip at an average speed of 45 km/h, he continued to travel another 240 km to reach Town Q.
   (a) Find the distance between the two towns.
   (b) If his average speed for the whole trip was 54 km/h, find his average speed for the last $\frac{2}{3}$ of the trip.